Puzzle RL Theory:
Where RL Seems To Matter Beyond SFT

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Scope

This note answers one narrow question: what reinforcement learning is buying over supervised fine-tuning in current LLM reasoning systems, and how to test that cleanly in a controlled puzzle setting. The target object is not "which method gets the highest benchmark number?" It is the mechanism split between:

- fixed-dataset imitation,
- search or sampling from the base policy,
- online data refresh from self-generated rollouts,
- sparse outcome-reward RL,
- and denser step-wise credit assignment.

Short answer

1. **Exact theory:** the clean objective bridge is fixed-reference KL-regularized RL. In that setting, maximizing expected reward minus a KL penalty is exactly equivalent to reverse-KL projection onto a Gibbs-reweighted target policy. This is the strongest safe "RL vs distillation" statement because it is an identity, not an analogy.
2. **Current empirical read:** much of ordinary token-space RLVR looks more like distribution sharpening than support expansion. Several supplied papers show that strong gains can often be recovered by better sampling, online self-training, or best-of-N filtering plus distillation.
3. **Where RL still seems to matter:** when correct full trajectories are extremely rare, plain outcome-only RLVR is too weak. In that regime, denser credit assignment or more diversity-preserving exploration may add signal that fixed-trace imitation and sparse RL both miss.

1. What is actually different about RL and SFT?

Fixed-data imitation versus moving-data optimization

SFT minimizes negative log-likelihood on a fixed dataset of prompt-response pairs. RL optimizes expected reward on trajectories sampled from the current policy. That means RL changes two things at once:

- the objective, because reward is not the same as token imitation;
- the training distribution, because the rollouts come from the current policy and therefore move during training.

This second difference is easy to understate. Several of the strongest non-RL baselines in the supplied set work surprisingly well precisely because they keep the data on-policy even without using a reward model.

The exact KL bridge

For prompt distribution $p(x)$, reward $r(x, y)$, reference policy $\pi_{\text{ref}}(y \mid x)$, and candidate policy $\pi(y \mid x)$, consider

$$J(\pi) = \mathbb{E}_x \left[\mathbb{E}_{y\sim\pi}[r(x, y)] - \beta \, \mathrm{KL}(\pi(\cdot \mid x) \,\|\, \pi_{\text{ref}}(\cdot \mid x))\right].$$

Then the optimizer is the Gibbs-reweighted policy

$$\pi^\star(y \mid x) \propto \pi_{\text{ref}}(y \mid x) \exp(r(x, y)/\beta),$$

and maximizing $J(\pi)$ is exactly equivalent to minimizing $\mathrm{KL}(\pi \,\|\, \pi^\star)$ up to an additive constant.

2. What the supplied papers say

2.1 Evidence that a lot of the gain is not unique to RL

Sampling alone can recover much of the gain. "Reasoning with Sampling" argues that a training-free sampler built from the base model's own likelihoods can nearly match or beat GRPO on several reasoning tasks while preserving pass@k and diversity.[1] The key point is not that training never helps. It is that a surprisingly large amount of post-RL performance seems to live in high-likelihood regions the base model already knows how to reach.

Reward-free online self-training can also recover much of the gain. "A Model Can Help Itself" shows that self-generated supervision alone can improve reasoning on some models without any external reward or verifier.[2] Two details are especially relevant for our project:

- **online refresh matters:** frozen self-generated data are much weaker than repeatedly refreshing with the latest policy;
- **temperature decoupling matters:** low-temperature generation plus standard-temperature training works far better than coupling the two temperatures.

So one part of what people call "RL gain" is often just the value of staying on-policy.

Best-of-N positives plus cloning already go far in binary-feedback settings. "OREAL" proves that, under binary outcome feedback, behavior cloning on positive trajectories from best-of-N sampling is sufficient for the KL-regularized optimal policy on the positive support.[3] Their added value comes from handling negative samples better and from adding token-level credit assignment, not from disproving the power of search-plus-distill.

[1] Source [4].
[2] Source [5].
[3] Source [2].

Standard RLVR often looks like trend amplification. "On the Linearity of LLMs' RLVR Training" finds strong approximate linearity in both weights and output log-probabilities over RLVR training steps.[4] Their interpretation is that RLVR often keeps amplifying directions found early, rather than continually discovering qualitatively new behavior.

Improved answers do not imply better traces. "Beyond Semantics" is especially important for our later maze experiments.[5] In a controlled puzzle-style setting, the paper shows:

- trace validity is not a reliable predictor of final correctness;
- corrupted traces can train almost as well as correct ones, and can generalize better;
- GRPO can improve answer quality without improving trace validity.

2.2 Where RL-like learning still seems to add something real

When pass@k is effectively zero, sparse RL is too weak. "Supervised Reinforcement Learning" targets the regime where the model almost never samples a fully correct trajectory, so outcome-only RLVR has no useful signal and plain SFT overfits long demonstrations.[6] Their dense step-wise similarity reward over expert actions is best read as evidence that *credit assignment* can be the real bottleneck.

Diversity-preserving exploration may matter. "Beyond Mode Elicitation" directly targets diversity collapse.[7] Their main claim is stronger than a pass@1 win: latent-space exploration improves both pass@1 and pass@k relative to discrete RL baselines. If that result generalizes, it is one of the better candidates for "RL is doing more than just sharpening one dominant mode."

3. Best current synthesis

What seems safely established

- The exact theoretical bridge is fixed-reference KL-RL, not a blanket equivalence between all RL and all SFT.
- A large fraction of current token-space RLVR gains can be explained by sampling, verifier filtering, and online data refresh over behaviors already latent in the base model.
- Plain final-answer gains do not by themselves show better process quality or support expansion.
- Dense or structured credit assignment matters when correct full trajectories are too rare for outcome-only RL to learn from.
- Diversity must be treated as part of the scientific object, not a side metric.

[4] Source [6].
[5] Source [3].
[6] Source [7].
[7] Source [8].

What seems plausible but not yet proved

- Some diversity-preserving RL variants may truly enlarge the set of successful reasoning trajectories rather than only reweighting one mode.
- Latent-space exploration may be a better optimizer family than discrete token RL when the main failure mode is entropy collapse.
- In long-horizon agentic settings, on-policy learning may matter more than it does in single-turn math or code.

What is still open

- How often does RL find successful trajectories that large-k base sampling could not have found?
- When RL helps, is the key ingredient reward optimization, on-policy refresh, verifier filtering, or dense credit assignment?
- How much of any RL gain compresses back into an ordinary supervised dataset?
- What should count as genuine "distribution expansion": lower base-policy likelihood, new algorithm family, stronger OOD transfer, or all three?

4. What this means for the maze repo

The existing repo already has the right seed object:

- graph generation and shortest-path verification,
- supervised pretrain and midtrain stages,
- path validity and path optimality metrics.

What it does *not* yet have is the mechanism surface we need. The first implementation order should be:

1. add multi-sample evaluation with pass@k, diversity logging, and base-logprob logging;
2. add best-of-N filtering plus behavior cloning;
3. add reward-free online self-training;
4. only then add sparse binary-outcome RLVR;
5. add a step-wise dense-credit baseline after that;
6. treat diversity-preserving RL as a later-stage extension.

5. Falsifiable puzzle hypotheses

1. **Mostly-sharpening hypothesis:** on the current maze family, best-of-N plus cloning and reward-free online self-training will recover most of the pass@1 gain that sparse RLVR gets.
2. **Credit-assignment hypothesis:** on harder maze regimes where base pass@k is near zero, dense step-wise feedback will beat sparse outcome-only RLVR.
3. **Diversity hypothesis:** standard discrete RLVR will tend to improve pass@1 at the cost of pass@k, while any truly better exploration method must avoid that trade.
4. **Compression hypothesis:** if a gain is mostly due to better trajectory discovery rather than optimizer-specific magic, much of it should survive distillation back into a supervised dataset.

Bottom line

The strongest current reading is not "RL does nothing" and not "RL creates brand-new reasoning by default." It is more specific:

> Current LLM RL often behaves like on-policy distribution shaping over capabilities that are already latent in the base model. The main exceptions appear when better data refresh, better verifier-filtered search, or better credit assignment materially change the learning problem. The maze project should be built to separate those ingredients one by one.

Sources

1. Internal bridge note: `projects/llm-rl-kl-duality/docs/theory-bridge.md`
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3. arXiv:2505.13775 – *Beyond Semantics: The Unreasonable Effectiveness of Reasonless Intermediate Tokens*
4. arXiv:2510.14901 – *Reasoning with Sampling: Your Base Model is Smarter Than You Think*
5. arXiv:2510.18814 – *A Model Can Help Itself: Reward-Free Self-Training for LLM Reasoning*
6. arXiv:2601.04537 – *Not All Steps are Informative: On the Linearity of LLMs' RLVR Training*
7. arXiv:2510.25992 – *Supervised Reinforcement Learning: From Expert Trajectories to Step-wise Reasoning*
8. arXiv:2602.01705 – *Beyond Mode Elicitation: Diversity-Preserving Reinforcement Learning via Latent Diffusion Reasoner*
9. arXiv:2507.10616 – *Scalpel vs. Hammer: GRPO Amplifies Existing Capabilities, SFT Replaces Them*